2025-2026 – Econ 0039 – Advanced Macro

Lecture 1: (Very) Long Run Growth

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Changes from version 1.0 are in red
(some figures were removed)

Disclaimer

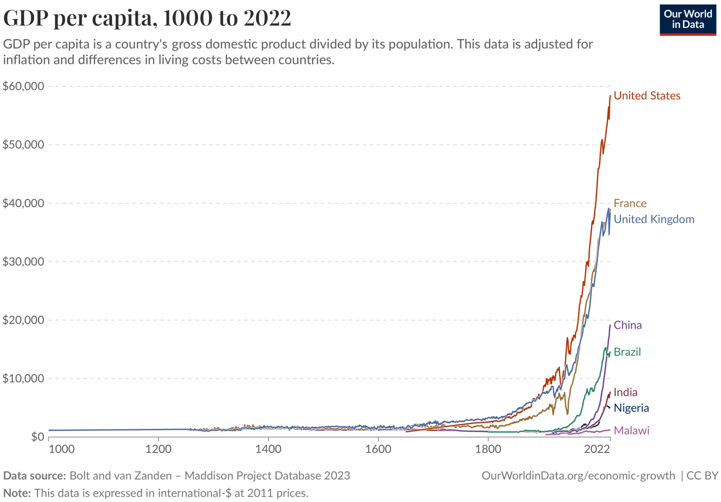
These are the slides I am using in class. They are not self-contained. They do contain original material and some that is not, including some "cut and paste" from various sources that I am citing as much as possible. These slides are not intended to be used outside of the course nor to be distributed. Thank you for signalling me typos or mistakes at f.portier@ucl.ac.uk.

- In this lecture, we study the very long run dynamics of economies.
- First we will identify three stages in economic history:
  - the Malthusian regime
  - The Industrial Revolution (and the demographic transition)
  - The Great Divergence
- Then we will use models to understand some of those evolutions (not the great Divergence)
  - A demographic model of the Malthusian regime
  - A fully specified general equilibrium model of Unified Growth Theory
- Compared to simple growth models, we will consider both economic and demographic trajectories of economies across time.

1. The Big Picture

Economic history of the world in one graph

Figure 1: Income per capita from 1000 to 2022 (ourworldindata.org)



1. Malthusian Regime (before 1760 in the UK): income per capita is roughly constant, close to subsistence level. Changes in the availability of resources translate into population changes
2. Industrial Revolution and the demographic transition (1760-1900 in the UK): rapid economic growth fuelled by increasing production efficiency + increase of population smaller than the increase of total income.
3. The Great Divergence (after 1900): some countries became unprecedentedly rich while standards of living declined or grew slowly in other.

2. The Matlhusian Regime

Reverend Thomas Robert Malthus (1766-1834)

- Professor of History and Political Economy at the East India Company College in Hertfordshire
- 1798: *An Essay on the Principle of Population*
- He argues that population growth generally expanded in times and in regions of plenty until the size of the population relative to the primary resources caused distress.

Figure 2: Malthus

2. The Matlhusian Regime

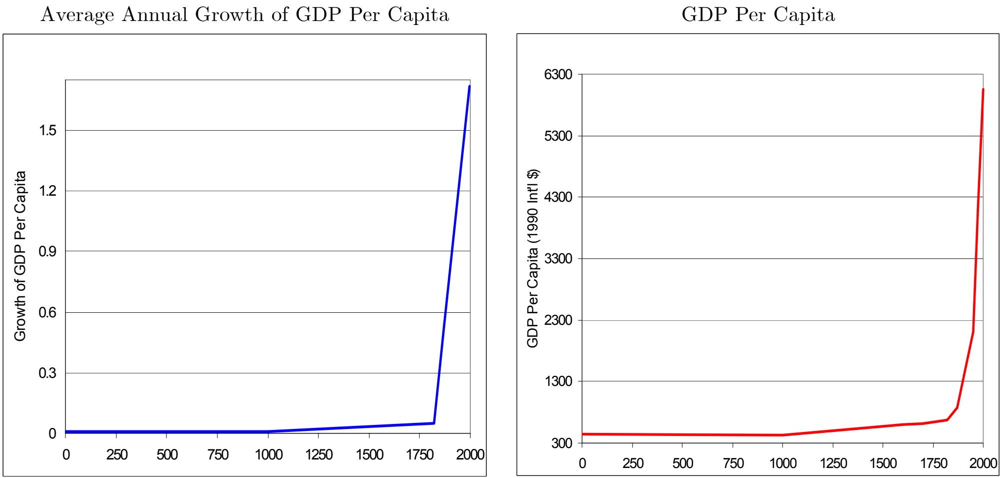
Reverend Thomas Robert Malthus (1766-1834)

- Two types of checks hold population within resource limits:
  - positive checks, which raise the death rate : *hunger, disease and war*.
  - preventive checks, which lower the birth rate : *abortion*, *birth control*, *prostitution*, *postponement of marriage and celibacy*.

- Humans struggled for life
- Technological progress was small
- Technological progress and land expansion caused an increase in the size of the population, not in the standards of living.
- Income per capita stayed in the proximity of a subsistence level.
- We'll see later in this lecture a formal model of the Malthusian regime

2. The Matlhusian Regime

Figure 3: Income per capita dynamics (Galor [2005])

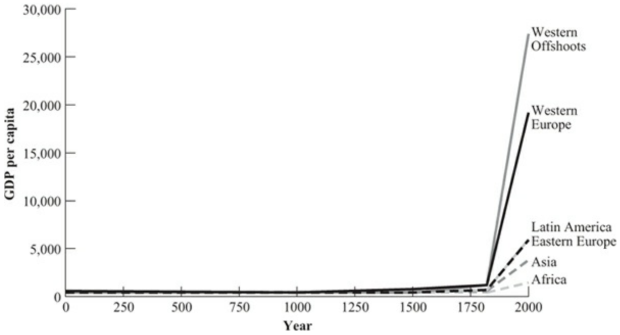


2. The Matlhusian Regime

Income per capita

- Average growth rate of income per capita is negligible : 0.05% a year between 1000 and 1820 (Maddison)
- Standards of living do not vary much across countries

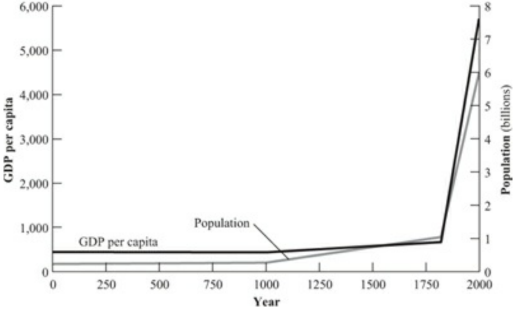
Figure 4: Across the world (Galor [2011])



Notes: in 1990 international $. Source: Maddison *[2003]*

- World population was 231 millions in 1 AD and 268 millions in 1000 AD, then 438 millions in 1500 AD
- Resource expansions after 1500 $\rightsquigarrow$ 1041 millions people in 1820

Figure 5: World population and income per capita from 1 AD to 2000 (Galor [2011])

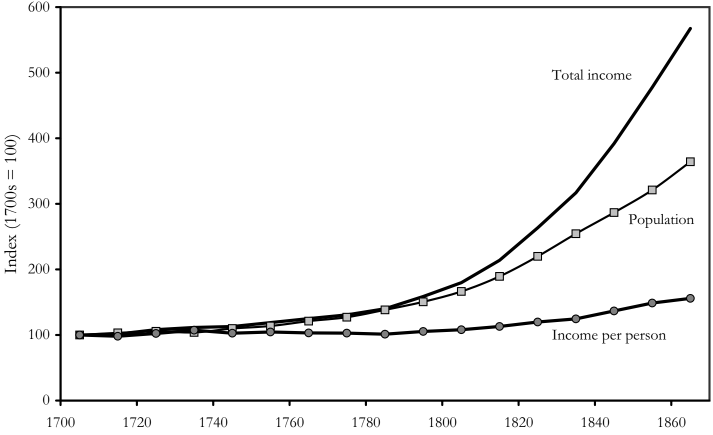


Notes: in 1990 international \$. Source: Maddison *[2003]*

2. The Matlhusian Regime

Population and income

Figure 6: Population and economic growth in England, 1700s-1860s (Clark [2007])



- English Black Death (1348-1349) : population went down from 6 millions to 3.5 millions
- Increase in the land/labor ratio $\rightsquigarrow$ the real wage tripled for 150 years
- This caused an increase in population $\rightsquigarrow$ in the 1560s, the real wage is back to the pre-plague level.

Figure 7: Population and Real Wages in England (Galor [2011])

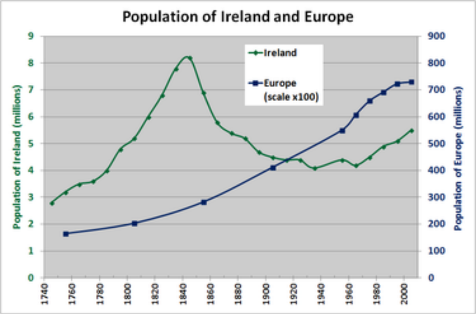
Source: Clark [2005]

2. The Matlhusian Regime

Example II: The Great Famine in Ireland (1845-1852)

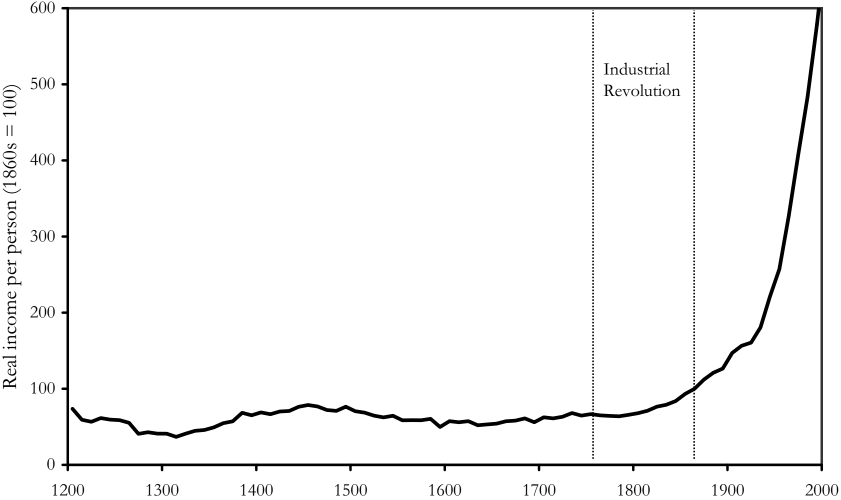
- Introduction of the potato in Ireland in the middle of the 17th century $\rightsquigarrow$ population increase
- Potato desease ("potato blight")
- About 1 million people died + massive emigration $\rightsquigarrow$ population decline.

Figure 8: Evolution of population and the Great Famine (WIKIPEDIA)



3. The Industrial Revolution

Figure 9: Real income per person in England 1260s-2000s (Clark [2007])



- It is a period of transition to new manufacturing processes
- Started in England around 1760 and went on up to 1840
- Spread over the European continent (Belgium Wallonia first) and to the U.S.A.
- The pace of technological progress increased with industrialization
- But it is also a time of great improvement in agricultural productivity (also an *agricultural revolution*)
- Simultaneous population and income per capita growth

- Textile : Mechanized cotton spinning powered by steam or water increased the output of a worker by a factor of about 1000.
- Steam power : The efficiency of steam engines increased so that they used between one-fifth and one-tenth as much fuel.
- Iron making : The substitution of coke (made by heating coal or oil in the absence of air) for charcoal greatly lowered the fuel cost of iron.

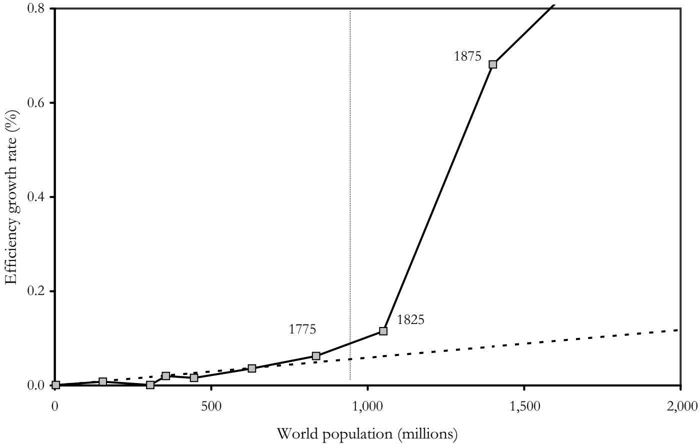
3. The Industrial Revolution

Figure 10: Real Wages in England and France During the take-off from the Malthusian Epoch (Galor [2005])

$$Y_t = A_t F(K_t, L_t) \Longleftrightarrow \frac{\Delta Y}{Y} = \underbrace{\frac{\Delta A}{A}}_{\text{Efficiency growth}} + \varepsilon_K^F \frac{\Delta K}{K} + \varepsilon_L^F \frac{\Delta L}{L}$$

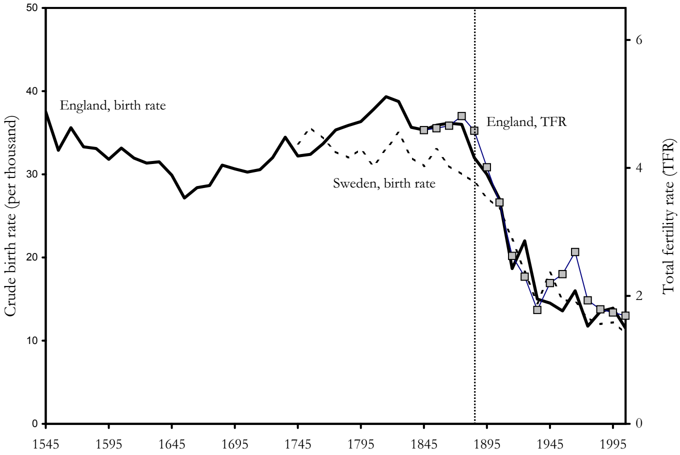
- ε_K^F: elasticity of F with respect to $K \approx 1/3$
- ε_L^F: elasticity of F with respect to $L \approx 2/3$
- A measures efficiency
- The *Solow residual* $\frac{\Delta A}{A}$ measures the growth in output that cannot be explained by the growth in inputs)

Figure 11: World population and growth rate of efficiency (Clark [2007])



- Let's have a look at two measures of birth rates.
- *Crude birth rate* indicates the number of live births occurring during the year, per 1000 population estimated at midyear.
- *Total fertility rate* is the average number of children that would be born to a woman over her lifetime if:
  - she were to experience the exact current age-specific fertility rates through her lifetime, and
  - she were to survive from birth through the end of her reproductive life.

Figure 12: Demographic transition (Clark [2007])



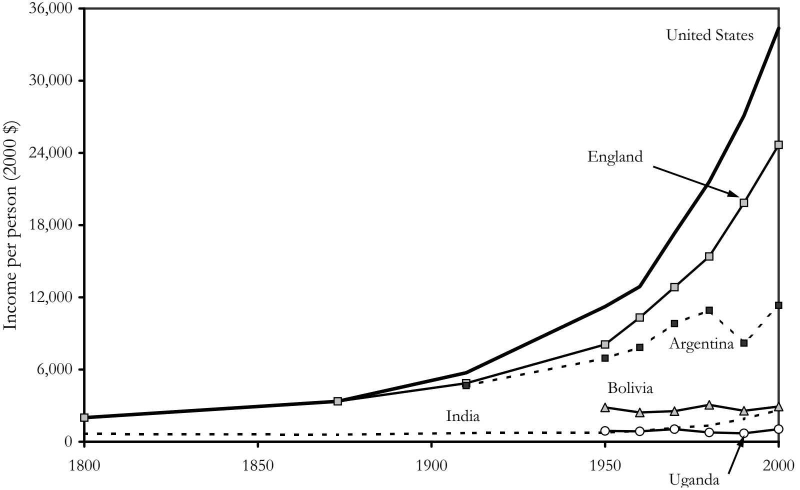
4. The Great Divergence

- Last two centuries : dramatic changes in the distribution of income and population across the globe.
- Inequality in the world economy was negligible untill the 19th century.

Table 1: Ratio of GDP per capita between the richest region and the poorest region in the world (Galor [2005])

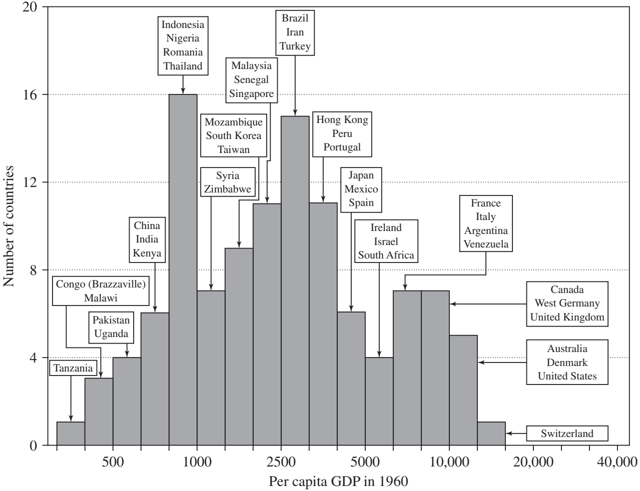
Year	1000	1500	1820	1870	1913	1950	2001
Ratio	1.1:1	2:1	3:1	5:1	9:1	15:1	18:1

Figure 13: Some countries took-off, some did not (CLARK [2007])



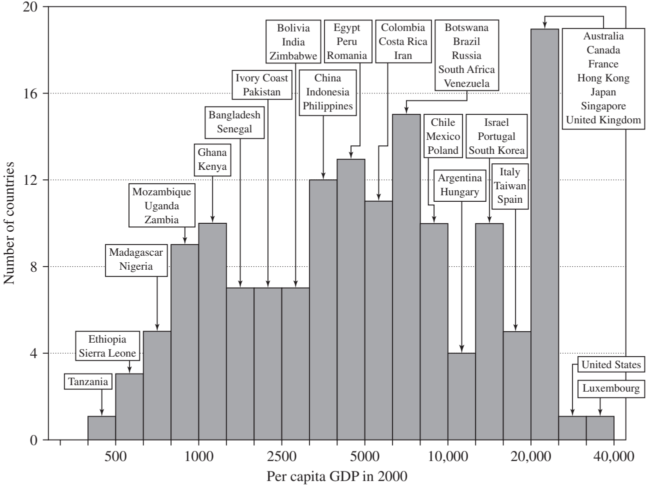
4. The Great Divergence

Figure 14: Inequalities across countries in 1960 (Barro & Sala i Martin [2004])



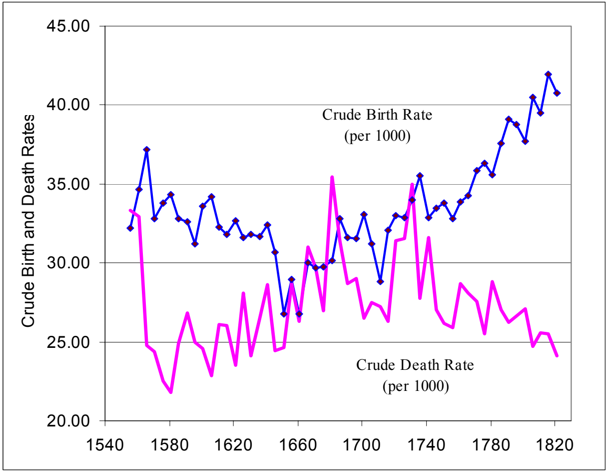
4. The Great Divergence

Figure 15: Inequalities across countries in 2000 (Barro & Sala i Martin [2004])



- Those facts raise a lot of questions:
  - × Why poor countries are not converging to the income per capita of the rich?
  - × If $Y = AK^{\alpha}L^{1-\alpha}$, why doesn't capital fly from *capital-abundant and low-marginal-productivity-of-capital* to *capital-poor and high-marginal-productivity-of-capital* ones?
  - × Marginal productivity of capital $= \alpha A \left(\frac{L}{K}\right)^{1-\alpha}$
  - × Are there non economic factors behind the non convergence (institutions? geography?)
- A lot of fascinating questions that we will not address in this lecture
- This is the field of *Growth Theory* and *Development Economics*

Figure 16: Crude birth and death rates in England (Galor [2005])



5. Demographics and the Simple Functioning of the Malthusian Regime

Some facts, Malthusian regime and after

Figure 17: Crude death rates (Galor [2005])

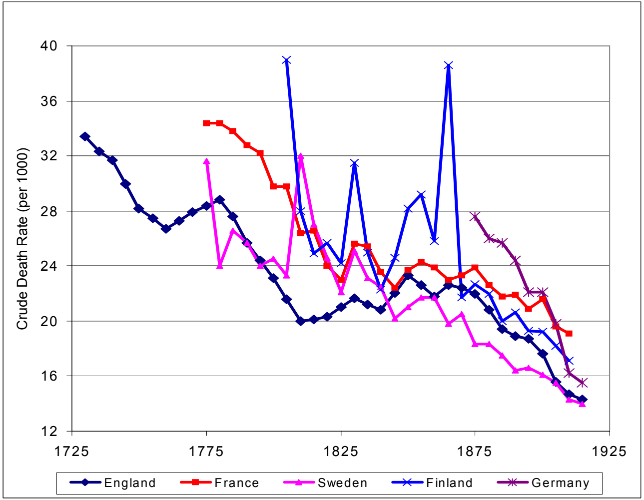
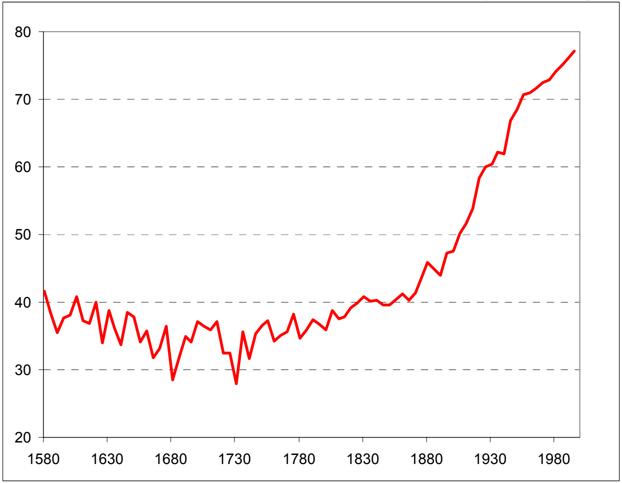


Figure 18: Life expectancy at birth in England (Galor [2005])



- See Problem VI in Problem set I for a micro-founded model (Ashraf and Galor, The American Economic Review, Vol. 101, No. 5, 2011)
- The model is adapted from Clark's book "Farewell to Alms".
- Time is continuous
- In the Mathusian regime,

  Demography (1): the birth rate b is high when income per capita y is high.

$$b(t) = b\underbrace{(y(t))}_{+}$$

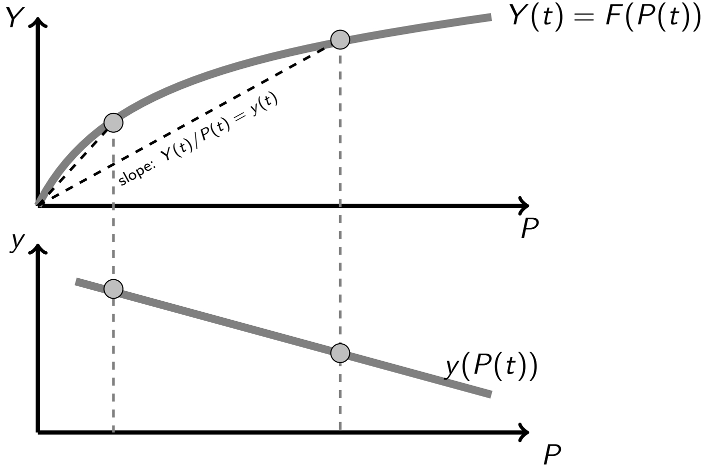
  Demography (2): the mortality rate m is low when income per capita y is high.

$$m(t) = m\underbrace{(y(t))}_{-}$$

  Technology: Population P is low when income per capita y is high.

$$P(t) = P\underbrace{(y(t))}_{-}$$

Figure 19: The technological schedule $P(t) = P(y(t))$ *(what is shown is the inverse function $y(t) = y(P(t))$)*

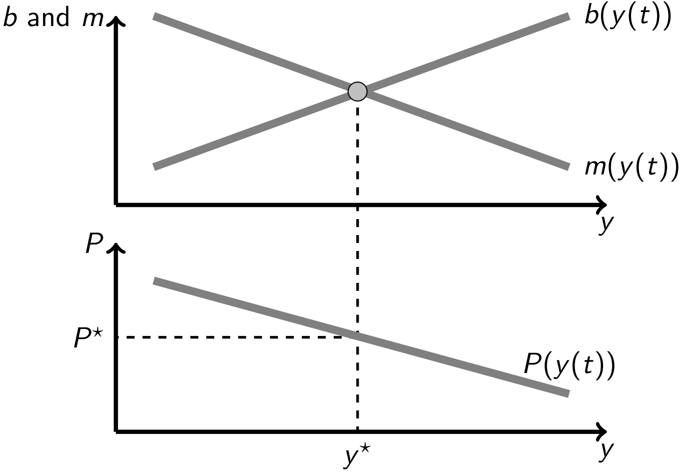


- All the dynamics of the model is given by the population law of motion

$$\dot{P}(t) = \Big(b(y(t)) - m(y(t))\Big)P(t)$$

- The steady state level of population $P^\star$ is such that $\dot{P}(t) = 0$.

Figure 20: Long run equilibrium determination



- All the dynamics of the model is given by the population law of motion:

$$\dot{P}(t) = \Big(b(y(t)) - m(y(t))\Big)P(t)$$

- Let's see what happens if the economy starts from a population $P(0)$ lower than the steady state one $P^{\star}$.

Figure 21: Dynamics starting from $P(0) < P^{\star}$

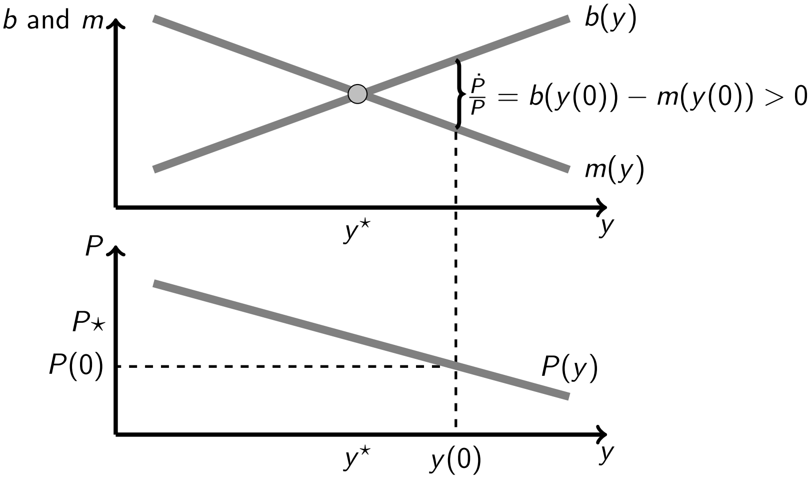


Figure 22: Dynamics starting from $P(0) < P^{\star}$

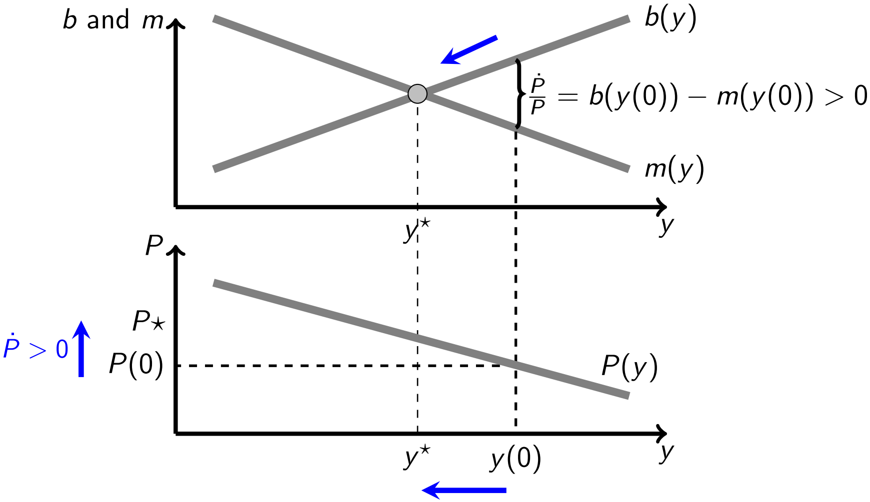


Figure 23: Dynamics starting from $P(0) < P^{\star}$

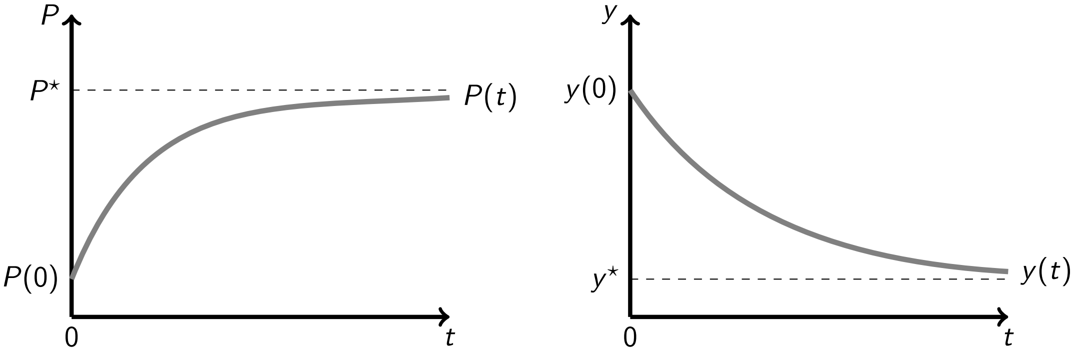


Figure 24: An increase in the birth rate

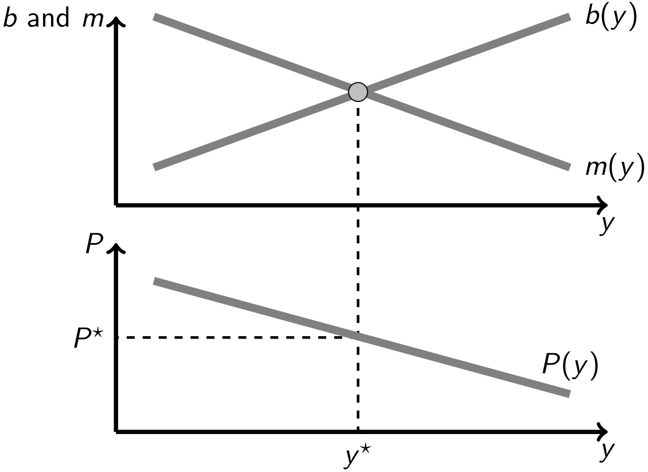


Figure 25: An increase in the birth rate

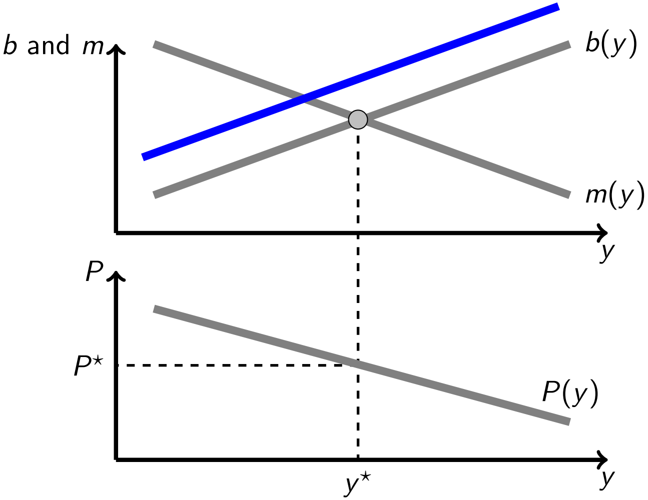


Figure 26: An increase in the birth rate

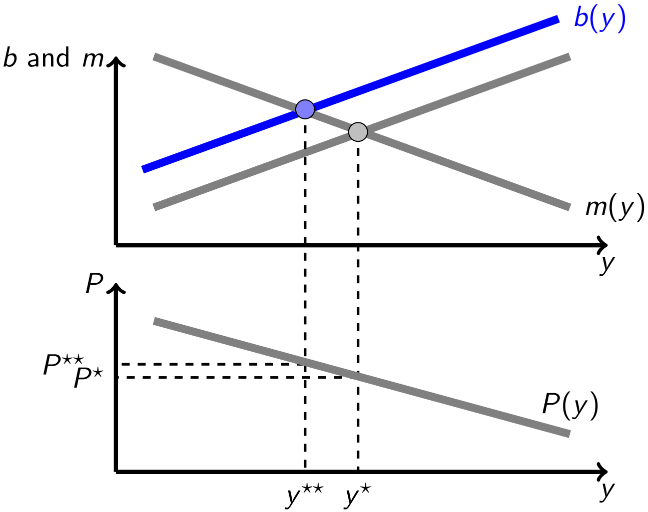


Figure 27: An increase in the birth rate

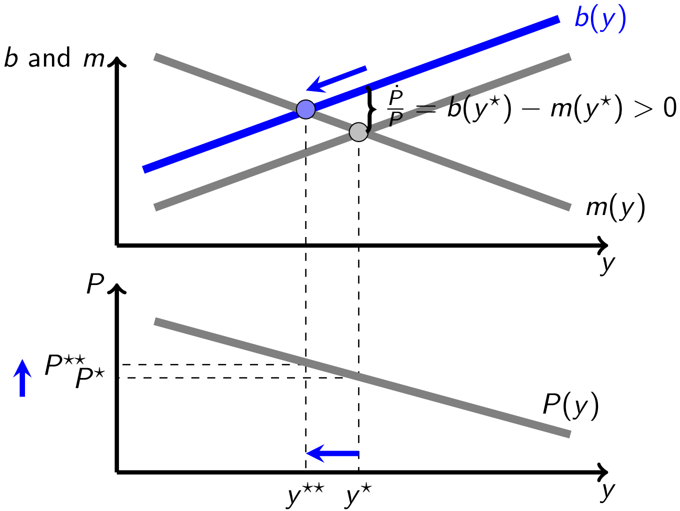


Figure 28: Dynamics following an increase in the birth rate

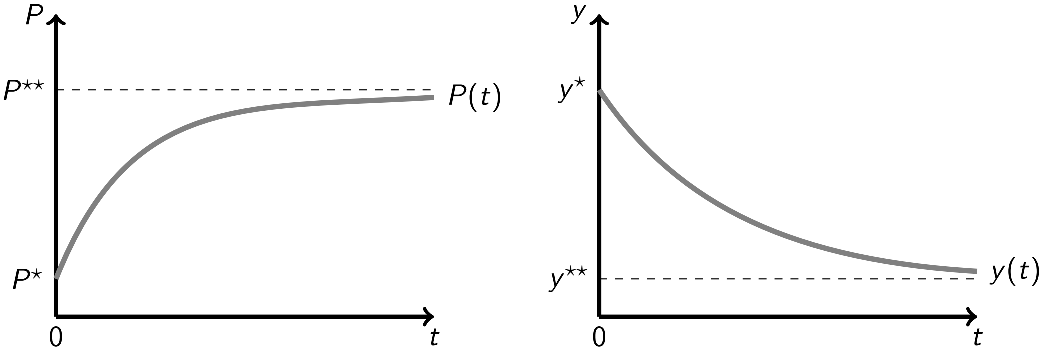


Figure 29: A decrease in the mortality rate

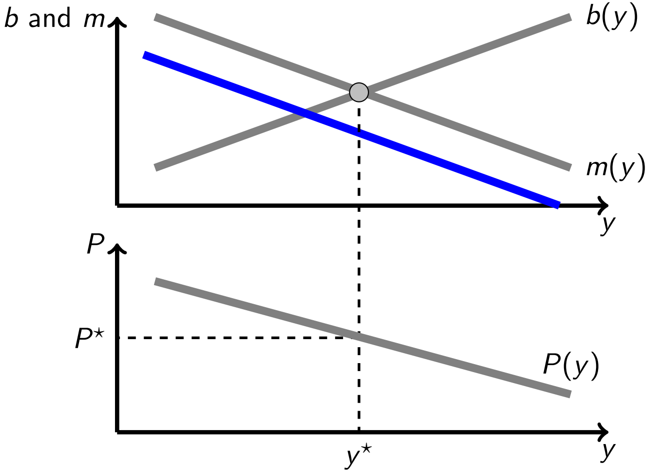
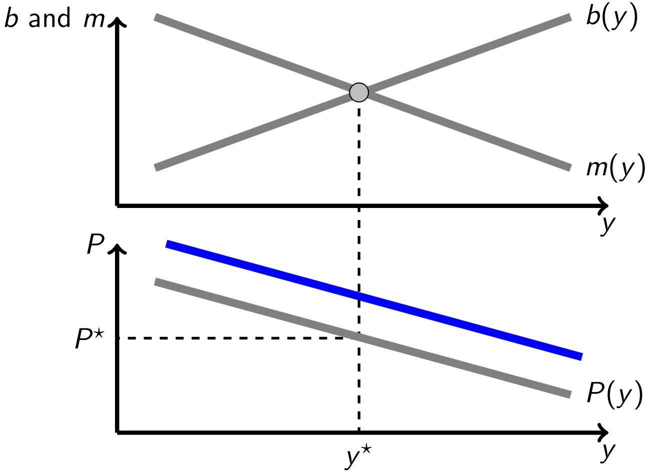


Figure 30: A technological progress



- Adapted from Strulik & Weisdorf [2008]
- This is a *Dynamic General Equilibrium model with micro foundations*.
- What does it mean?
  - Agents maximize utility under budget constraint
  - They live more than one period
  - Prices are determined by equality of demand and supply on all markets

- Agents live 2 periods (OLG (*OverLapping Generations*) model).
- childhood : consume, at the expense of the parents.
- adulthood: work 1 unit, have children, feed them, consume, and then die.
- In a given period, population is composed of children and adults.
- The period after, adults are dead, children are adults and they have children.
- Simplifying assumption: asexual reproduction: one (single) adult has n_t children.
- Labor force dynamics : $L_{t+1} = n_t L_t$
- n_t = fertility (no children die)
- All agents are identical

- Adults like manufacturing goods m_t and children n_t.
- quasi-linear utility : $u_t = m_t + \gamma \log n_t$.
- Children like only food.
- Price of m is 1 by normalization, price of food is p.
- For simplicity, people consume food only when young. They need $a = 1$ unit of food.
- Therefore, 1 child costs $a = 1$ unit of food.
- The budget constraint of an adult is therefore $p_t n_t + m_t = w_t$
- w_t is an adult income

$$\max_{n_t, m_t} \quad m_t + \gamma \log n_t$$
$$s.t. \quad p_t n_t + m_t = w_t$$

- solution : $\max_{n_t} w_t - p_t n_t + \gamma \log n_t \rightsquigarrow$ First Order Condition : $n_t = \frac{\gamma}{p_t}$
- Discussion
  - × Strong assumption on preferences for children: income does not affect n (zero income elasticity)
  - × Only the relative price of food affect fertility decisions.
  - × But there is an indirect effect of growth (and therefore income) on fertility:
    - Agric. productivity growth $\rightsquigarrow$ p decreases $\rightsquigarrow$ fertility increases.
    - Manuf. productivity growth $\rightsquigarrow$ p increases $\rightsquigarrow$ fertility decreases.
- Malthus preventive check hypothesis : relatively scarce agricultural goods (high p) $\rightsquigarrow$ low fertility.

- All technologies are constant returns.
- Food: $Y_t^A = A_t^{\epsilon}(L_t^A)^{\alpha} X^{1-\alpha} \quad , \ \alpha \in]0,1[$
- X is land, assumed to be in fixed supply, and normalized to $X = 1$; L^A is agricultural labor and A is productivity on agriculture. ϵ is between 0 and 1.
- Manufacturing: $Y_t^M = M_t^{\epsilon} L_t^M$
- Assume for simplicity that land is free disposal (land rent is zero)

- Manufacturing sector
  - The (representative) competitive manufacturing firm will maximise profit $\Pi_t = Y_t^M - w_t^M L_t^M = M_t^\epsilon L_t^M - w_t^M L_t^M = \left(M_t^\epsilon - w_t^M\right) L_t^M$.
  - Optimal production will be either zero or infinity if $w_t^M \neq M_t^\epsilon$. This cannot be an equilibrium.
  - Therefore, one must have $w_t^M = M_t^\epsilon$
  - Note that, because of constant returns and competitive behaviour, $\Pi_t = 0$
- Food sector
  - In the food sector, we have assumed that land is free disposal.
  - We also assume that food production is individually done by each individual, and that the worker receives all the revenues from production
  - Therefore, the wage in the agricultural sector is the average productivity of labor: $w_t^A = \frac{p_t Y_t^A}{L_t^A} = p_t A_t^\epsilon (L_t^A)^{\alpha-1}$

- Growth is endogenous, and comes from technological progress $A_{t+1} - A_t$ and $M_{t+1} - M_t$.
- We assume *Learning-by-doing*:
  - $A_{t+1} - A_t = Y_t^A$
  - $M_{t+1} - M_t = Y_t^M$
- Note that this comes as an externality: no one has a patent for the improvement in technology, i.e. for the increase in A and M.
- No capital for simplicity.
- Notation: for a variable Z, $g_t^Z = \frac{Z_{t+1} - Z_t}{Z_t}$
- Therefore,

$$\begin{aligned} g_t^A &= \frac{A_{t+1} - A_t}{A_t} &= \frac{Y_t^A}{A_t} &= A_t^{\epsilon - 1}(L_t^A)^\alpha \\ g_t^M &= \frac{M_{t+1} - M_t}{M_t} &= \frac{Y_t^M}{M_t} &= M_t^{\epsilon - 1} L_t^M \end{aligned}$$

- What is an equilibrium in such an economy?
- In period t, given L_t and productivities A_t and M_t:
  - × Adults work 1 unit and choose optimally fertility (=food demand) and manuf. goods demand given prices and their income.
  - × Adults choose to work as farmers or manufacturers.
  - × Labor market clears.
  - × Food and manufacturing goods markets clear.
- Solving for the general equilibrium means to find p_t, w_t, n_t, Y_t^A, Y_t^M.
- Once we have solved, we can compute L_{t+1} and productivities A_{t+1} and M_{t+1} $\rightsquigarrow$we can move on and solve for period $t+1$
- Once we are given L_0, A_0 and M_0, we can solve for the whole history of human activity: everything is endogenous: production, population and technology.
- This is a model of *Unified Growth*.

- Equilibrium on the food market:
  - demand: $n_t L_t$
  - supply : $A_t^{\epsilon}(L_t^A)^{\alpha}$
  - Let $\theta_t = \frac{L_t^A}{L_t}$ be the share of total labor devoted to agriculture, food supply = food demand implies

$$\theta_t = \left(\frac{n_t L_t^{1-\alpha}}{A_t^{\epsilon}}\right)^{\frac{1}{\alpha}}$$

  - Details: $n_t L_t = A_t^{\epsilon}(L_t^A)^{\alpha}$. Replace L_t^A by $\theta_t L_t$ and rearrange terms.

- Equilibrium on the labor market:
  - Individuals freely choose in which sector to work.
  - The wage has to be the same in the two sectors for production to be non zero in both sectors:
$$w_t^A = w_t^M$$
  - Using the expressions for w_t^A and w_t^M previously computed, we obtain (see next slide)
$$p_t = \frac{M_t^{\alpha\epsilon}(\gamma L_t)^{1-\alpha}}{A_t^{\epsilon}}$$
  - Given that $p_t = \frac{\gamma}{n_t}$, we obtain
$$n_t = \gamma^{\alpha} \frac{A_t^{\epsilon}}{M_t^{\alpha\epsilon} L_t^{1-\alpha}}$$

- Take $w_t^A = w_t^M$ and replace the wages by their expression to obtain $\frac{p_t Y_t^A}{L_t^A} = \frac{Y_t^M}{L_t^M}$.
- Replace Y_t^A by $A_t^\epsilon (L_t^A)^\alpha$ and Y_t^M by $M_t^\epsilon L_t^M$ to obtain $\frac{p_t A_t^\epsilon (L_t^A)^\alpha}{L_t^A} = \frac{M_t^\epsilon L_t^M}{L_t^M}$
- Now replace L_t^A by $\theta_t L_t$, θ_t by $\left(\frac{n_t L_t^{1-\alpha}}{A_t^\epsilon}\right)^{\frac{1}{\alpha}}$ and n_t by $\frac{\gamma}{p_t}$ to obtain

$$p_t = \frac{M_t^{\alpha\epsilon}(\gamma L_t)^{1-\alpha}}{A_t^\epsilon}$$

- Given that $p_t = \frac{\gamma}{n_t}$, the previous equation rewrites

$$n_t = \gamma^\alpha \frac{A_t^\epsilon}{M_t^{\alpha\epsilon} L_t^{1-\alpha}}$$

- The model dynamics is then computed as follows:
- Take an initial period 0. (L_0, A_0, M_0) are given.
- Fertility (and therefore population growth) is given by

$$n_0 = \gamma^\alpha \frac{A_0^\epsilon}{M_0^{\alpha\epsilon} L_0^{1-\alpha}}$$

- The equilibrium allocation of labor between agriculture and manufacturing is then given by

$$\theta_0 = \left(\frac{n_0 L_0^{1-\alpha}}{A_0^\epsilon} \right)^{\frac{1}{\alpha}}$$

so that $L_0^A = \theta_0 L_0$ and $L_0^M = (1 - \theta_0) L_0$

- Then production in each sectors are given by $Y_0^A = A_0^\epsilon (L_0^A)^\alpha$ and $Y_0^M = M_0^\epsilon L_0^M$.
- Form the learning-by-doing mechanism, we obtain the period 1 level of productivity: $A_1 = A_0 + Y_0^A$ and $M_1 = M_0 + Y_0^M$
- Finally, $L_1 = n_0 L_0$
- We have computed $(L_1, A_1, M_1) \rightsquigarrow$ we can now move on and compute the other equilibrium variables of period 1, etc...

- BGP are paths along which variables grow at constant rate (possibly zero).
- θ must be constant along a BGP, meaning that L^A and L^M grow at the same rate, that we denote g^L.
- We have shown that $g_t^A = A_t^{\epsilon-1}(L_t^A)^\alpha$. Along a BGP, g_t^A is constant, implying that
$$A_t^{\epsilon-1}(L_t^A)^\alpha = g^A \text{ and } A_{t+1}^{\epsilon-1}(L_{t+1}^A)^\alpha = g^A$$
- Taking the ratio,
$$\left(\frac{A_{t+1}}{A_t}\right)^{\epsilon-1}\left(\frac{L_{t+1}^A}{L_t^A}\right)^\alpha = 1,$$
which implies $\left(1+g^A\right)^{\epsilon-1}\left(1+g^L\right)^\alpha = 1 \iff (1+g_A) = (1+g_L)^{\alpha/(1-\epsilon)}$.
- Similarly, $g_t^M = M_t^{\epsilon-1}L_t^M$ implies $(1+g_M) = (1+g_L)^{1/(1-\epsilon)}$.
- Therefore, $(1+g_M)^\alpha = (1+g_A)$.

- From

$$n_t = \gamma^{\alpha} \frac{A_t^{\epsilon}}{M_t^{\alpha\epsilon} L_t^{1-\alpha}},$$

given that $n_t = (1 + g_t^L)$, along a BGP $\frac{n_{t+1}}{n_t} = 1$,so that

$$\frac{n_{t+1}}{n_t} = 1 = \frac{(A_{t+1}/A_t)^{\epsilon}}{(M_{t+1}/M_t)^{\alpha\epsilon}(L_{t+1}/L_t)^{1-\alpha}} = \frac{(1+g^A)^{\epsilon}}{(1+g^M)^{\alpha\epsilon}(\underbrace{n}_{1+g^L})^{1-\alpha}}$$

- Using $(1 + g_M)^{\alpha} = (1 + g_A)$, this equation becomes :

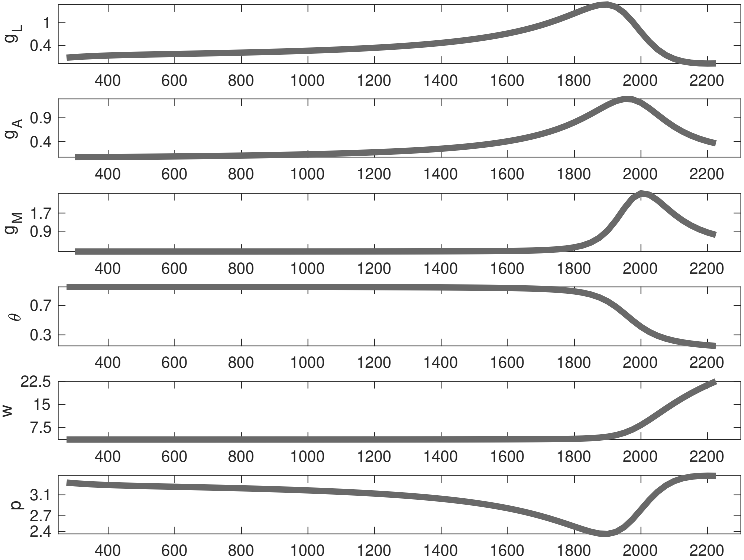
$$(1 + g^L)^{\alpha-1} = 1 \iff g^L = 0$$

- Population is constant in the long run.
- If $g^L = 0$ along a BGP, then g_M and g_A are also zero $\rightsquigarrow$ the economy will eventually reach a stationary state with zero growth.

- We can use a spreadsheet to compute the full dynamics of the economy starting from some initial conditions and for some values of the parameters.
- For this purpose, we need to slightly modify the model (theoretically unimportant, but useful for the simulations)
- $Y_t^A = A_t^\epsilon (L_t^A)^\alpha X^{1-\alpha} \rightsquigarrow Y_t^A = \mu A_t^\epsilon (L_t^A)^\alpha X^{1-\alpha}$
- $Y_t^M = M_t^\epsilon L_t^M \rightsquigarrow Y_t^M = \delta M_t^\phi L_t^M$
- In that case, it is easy to show that $n_t = \mu \left(\frac{\gamma}{\delta}\right)^\alpha \frac{A_t^\epsilon}{M_t^{\alpha\phi} L_t^{1-\alpha}}$ and that $\theta_t = \left(\frac{n_t L_t^{1-\alpha}}{\mu A_t^\epsilon}\right)^{\frac{1}{\alpha}}$
- Parameters: $\alpha = 0.8$, $\epsilon = 0.45$, $\phi = 0.3$, $\mu = 0.5$, $\delta = 1.5$, $\gamma = 3.4$ and initial values are $A_0 = 0.8$, $M_0 = 18.15$ and $L_0 = 0.014$.

6. A Model of the (Very) Long Economic History

An numerical example of the full dynamics

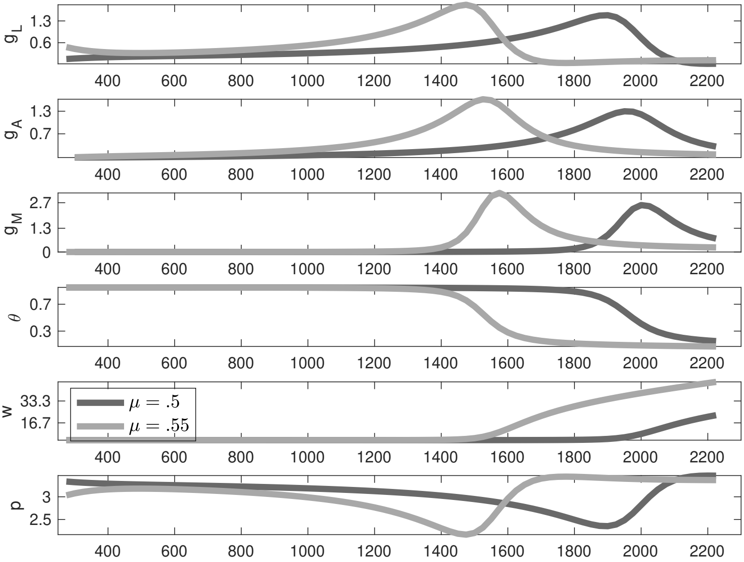


- Assume we choose a higher value of μ (0.55 instead of 0.5)

$$Y_t^A = \mu A_t^{\epsilon} (L_t^A)^{\alpha} X^{1-\alpha}$$

A simple model of the (very) long economic history

Increasing μ



7. Summary

- We have sketched the three phases of the economic history in the world: The Malthusian regime, the Industrial Revolution and the Great Divergence
- We have studied a demographic model of the Malthusian regime
- We have studied a dynamic general equilibrium model of Unified Growth Theory.
- This model endogenizes demographics and technical progress

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